

■ Plastic Recycling Guide

Comprehensive Environmental Guide

■ **Quick Tip:** Always rinse food containers before recycling — contaminated plastics are often rejected at sorting facilities and end up in landfill.

■ Definition

Plastic recycling is the process of collecting, sorting, cleaning, and reprocessing plastic waste materials into new products, thereby diverting them from landfills, oceans, and incinerators. Plastics are classified by resin identification codes (1–7) that indicate their chemical composition and recyclability.

Common recyclable plastics include PET (code 1, beverage bottles), HDPE (code 2, milk jugs, detergent bottles), and PP (code 5, yoghurt tubs, straws). Plastics coded 3, 6, and 7 are generally not accepted by kerbside recycling programmes due to complex chemistry or low market demand.

■■ Disposal Methods

- Kerbside/curbside collection: place accepted plastic types in designated recycling bins according to local guidelines.
- Drop-off recycling centres for plastics not accepted at kerb — particularly films, bags, and polystyrene.
- Supermarket take-back schemes for plastic bags and flexible film packaging.
- Chemical recycling (pyrolysis, depolymerisation) converts mixed plastics back into fuel or chemical feedstocks.
- Mechanical recycling: shredding, melting, and re-pelletising plastics into raw material for manufacturers.

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■ Environmental Impact

Plastics in the environment fragment into micro- and nanoplastics that contaminate soil, freshwater, and marine ecosystems. Microplastics have been detected in human blood, breast milk, and major organs. Ocean plastic pollution endangers marine life through ingestion and entanglement, with an estimated 8 million metric tonnes entering the sea each year.

Recycling plastic reduces the need for virgin fossil-fuel-derived resins, cutting greenhouse gas emissions and energy consumption by up to 70% compared with primary production. However, most plastic can only be mechanically recycled a limited number of times before quality degrades, making reduction and reuse essential complements to recycling.

■ Safety Guidelines

- Never burn plastic — combustion releases dioxins, furans, and other toxic compounds into the air.
- Do not recycle plastic bags or films in kerbside bins unless explicitly allowed; they jam sorting machinery.
- Rinse containers thoroughly to remove food residue before placing in the recycling bin.
- Flatten plastic bottles to save space, but leave caps on (most facilities now accept them together).
- Check local recycling codes — accepted materials vary significantly between municipalities.

■ Best Practices

- Follow the 3Rs hierarchy: Reduce consumption first, Reuse containers and bags, then Recycle as a last resort.
- Choose products with minimal or recyclable packaging; avoid single-use plastics whenever possible.
- Learn your local resin code acceptance list and label bins accordingly.
- Support businesses and policies that invest in extended producer responsibility (EPR) schemes.
- Participate in local clean-up events to prevent plastic from entering waterways.